

5.3 Find the minimum sum of products for each function using a Karnaugh map.

(a) $f_1(a, b, c) = m_0 + m_2 + m_5 + m_6$

(b) $f_2(d, e, f) = \Sigma m(0, 1, 2, 4)$

(c) $f_3(r, s, t) = rt' + r's' + r's$

(d) $f_4(x, y, z) = M_0 \cdot M_5$

- 5.4 (a) Plot the following function on a Karnaugh map. (Do not expand to minterm form before plotting.)

$$F(A, B, C, D) = BD' + B'CD + ABC + ABC'D + B'D'$$

- (b) Find the minimum sum of products.

- (c) Find the minimum product of sums.

5.5 A switching circuit has two control inputs (C_1 and C_2), two data inputs (X_1 and X_2), and one output (Z). The circuit performs one of the logic operations AND, OR, EQU (equivalence), or XOR (exclusive OR) on the two data inputs. The function performed depends on the control inputs:

C_1	C_2	Function Performed by Circuit
0	0	OR
0	1	XOR
1	0	AND
1	1	EQU

(a) Derive a truth table for Z .

(b) Use a Karnaugh map to find a minimum AND-OR gate circuit to realize Z .

5.6 Find the minimum sum-of-products expression for each function. Underline the essential prime implicants in your answer and tell which minterm makes each one essential.

(a) $f(a, b, c, d) = \Sigma m(0, 1, 3, 5, 6, 7, 11, 12, 14)$

(b) $f(a, b, c, d) = \Pi M(1, 9, 11, 12, 14)$

(c) $f(a, b, c, d) = \Pi M(5, 7, 13, 14, 15) \cdot \Pi D(1, 2, 3, 9)$

5.7 Find the minimum sum-of-products expression for each function.

(a) $f(a, b, c, d) = \Sigma m(0, 2, 3, 4, 7, 8, 14)$

(b) $f(a, b, c, d) = \Sigma m(1, 2, 4, 15) + \Sigma d(0, 3, 14)$

(c) $f(a, b, c, d) = \Pi M(1, 2, 3, 4, 9, 15)$

(d) $f(a, b, c, d) = \Pi M(0, 2, 4, 6, 8) \cdot \Pi D(1, 12, 9, 15)$

5.8 Find the minimum sum of products and the minimum product of sums for each function:

(a) $f(a,b,c,d) = \Pi M(0,1,6,8,11,12) \cdot \Pi D(3,7,14,15)$

(b) $f(a,b,c,d) = \Sigma m(1,3,4,11) + \Sigma d(2,7,8,12,14,15)$

5.9 Find the minimum sum of products and the minimum product of sums for each function:

(a) $F(A, B, C, D, E) = \Sigma m(0, 1, 2, 6, 7, 9, 10, 15, 16, 18, 20, 21, 27, 30)$
 $+ \Sigma d(3, 4, 11, 12, 19)$

(b) $F(A, B, C, D, E) = \Pi M(0, 3, 6, 9, 11, 19, 20, 24, 25, 26, 27, 28, 29, 30)$
 $\cdot \Pi D(1, 2, 12, 13)$

5.10 $F(a, b, c, d, e) = \Sigma m(0, 3, 4, 5, 6, 7, 8, 12, 13, 14, 16, 21, 23, 24, 29, 31)$

- (a) Find the essential prime implicants using a Karnaugh map, and indicate why each one of the chosen prime implicants is essential (there are four essential prime implicants).

- (b) Find all of the prime implicants by using the Karnaugh map. (There are nine in all.)

5.11 Find a minimum product-of-sums solution for f . Underline the essential prime implicants.

$$f(a, b, c, d, e) = \Sigma m(2, 4, 5, 6, 7, 8, 10, 12, 14, 16, 19, 27, 28, 29, 31) + \Sigma d(1, 30)$$

5.12 Given $F = AB'D' + A'B + A'C + CD$.

(a) Use a Karnaugh map to find the maxterm expression for F (express your answer in both decimal and algebraic notation).

(b) Use a Karnaugh map to find the minimum sum-of-products form for F' .

(c) Find the minimum product of sums for F .

- 5.13** Find the minimum sum of products for the given expression. Then, make minterm 5 a don't-care term and verify that the minimum sum of products is unchanged. Now, start again with the original expression and find each minterm which could *individually* be made a don't-care without changing the minimum sum of products.

$$F(A, B, C, D) = A'C' + B'C + ACD' + BC'D$$